



Features of special number sequences

Task A: Types of special number sequences

1) Fill in the missing numbers in these special number sequences.

triangular numbers	square numbers	cube numbers
1, 3, 6, 10, 15, 21, 28, 36, ...	1, 4, 9, 16, 25, 36, 49, 64, ...	1, 8, 27, 64, 125, 216, ...

2) Calculate the following:

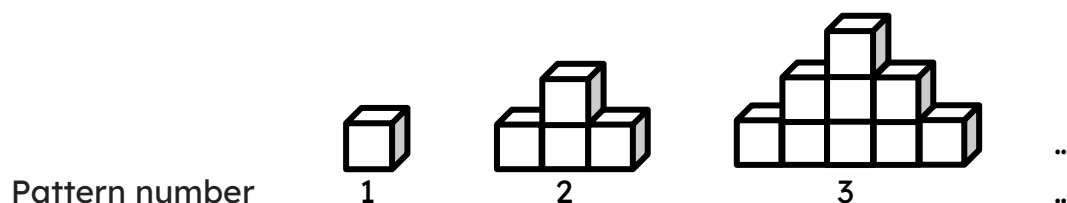
- a) The 5th square number plus the 5th **triangular number** 40
- b) The 8th **triangular number** plus the 4th cube number 100
- c) The 1st **triangular number** plus the 1st cube number 1
- d) The 3rd square number minus the 2nd cube number 1
- e) The 10th cube number minus the 10th square number 900
- f) The 11th **triangular number** plus the 12th **triangular number** 144

3) Sofia is having fun building different sequences using the square, **triangular** and cube numbers. Suggest what the next number in each sequence could be, explaining your answer.

- a) 81, 64, 49, 36, ... 25 - square numbers descending from 9^2
- b) 1, 6, 15, 28, ... 45 - every other **triangular number** starting with 1
- c) 2, 7, 15, 26, 40, ... 57 - square numbers plus **triangular numbers**
- d) 0, 4, 18, 48, 100, ... 180 - cube numbers minus square numbers
- e) 1, 1, 3, 4, 6, 9, 10, 16, ... 15 - alternating between **triangular** and square numbers
- f) 0, 1, 3, 6, 10, 15, 21, ... 28 - square numbers minus **triangular numbers**

Task B: Identifying special number sequences

1) Izzy is building step patterns out of cubes. Here are her first 3 patterns.



a) Fill in the table to show the number of cubes in her first 5 patterns if she continues to build them in the same way.

Pattern number	1	2	3	4	5
Number of cubes	1	4	9	16	25

b) What do you notice about your answers?

They are the square numbers

c) How many cubes would she need to build the 10th pattern? $10^2 = 100$

2) Alex is playing around with number patterns. He notices that if he squares the number 3, then subtracts one, then divides by 8 he gets a square number. He wants to see if this works for other odd numbers.

a) Fill in the table with the results from the first 5 odd numbers greater than 1

Start number (n)	3	5	7	9	11
Result ($\frac{n^2 - 1}{8}$)	1	3	6	10	15

b) What do you notice about your answers?

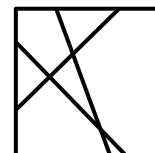
*They are the **triangular** numbers not the square numbers*

c) What value would you get if you applied this calculation to the number 19?

*19 is the 9th odd number greater than 1. The 9th **triangular** number is 45.*

3) Sam is investigating the maximum number of intersection points between straight lines drawn in a square. They want to know how many intersection points there could be if they drew 12 different lines, but drawing them all is proving difficult.

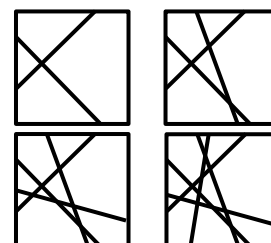
a) Sam tries it with 3 lines first, making sure the lines intersect the maximum number of times. How many intersection points are there? 3



b) Fill in the table for 2 lines, 3 lines, 4 lines and 5 lines, what do you notice about your answers?

Number of lines	2	3	4	5
Number of intersections	1	3	6	10

They are the **triangular** numbers.



c) What is the maximum number of intersections with 12 lines?

2 lines was the first **triangular** number.

3 lines the second **triangular** number.

So 12 lines should be the 11th **triangular** number which is

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 = 66$$

